

1. Administrative & Study Identification

Full Study Title: Study of Pembrolizumab (MK-3475) and Pembrolizumab With Other Investigational Agents in Participants With High Risk Non-muscle Invasive Bladder Cancer (MK-3475-057/KEYNOTE-057) **Short Title/Acronym:** KEYNOTE-057 **Protocol Number:** MK-3475-057 **NCT Number:** NCT02625961 **Sponsor Name:** Merck Sharp & Dohme Corp. (Merck) **Investigational Product:** Pembrolizumab (MK-3475) **Phase of Development:** Phase II **Trial Status:** Active but not currently recruiting (ACTIVE_NOT_RECRUITING) **Medical Monitor:** Sponsor Medical Monitor (Merck Clinical Director)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the Complete Response Rate (CRR) for patients with Carcinoma in situ (CIS) (Cohort A) and the 12-month Disease-Free Survival (DFS) rate for patients with papillary tumors without CIS (Cohort B) treated with pembrolizumab therapy or pembrolizumab in combination with other investigational agents. **Secondary Objectives:** To evaluate the CR Rate of any disease, CR Rate of High-Risk NMIBC in PD-L1 positive participants, and Duration of Response (DOR).

2.2 Background & Rationale Patients with high-risk NMIBC unresponsive to BCG therapy who are ineligible for or elect not to undergo radical cystectomy have a significant unmet medical need. Activation of the PD-1 pathway is implicated in BCG resistance, providing a strong rationale for evaluating the PD-1 inhibitor pembrolizumab as a viable bladder-sparing option. The primary study hypothesis is that treatment with pembrolizumab will result in a clinically meaningful response.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Multicohort design. *Note: Effective as of Amendment 12, Cohort C was made exploratory, and all Cohort C outcome measures were made exploratory.*)

3.2 Planned Enrollment & Duration Target **NS:** 320 patients **Number of Sites:** Multicenter (54 sites across 14 countries) **Estimated Study Start:** December 2015 **Estimated Primary Completion:** October 2022 (Data cutoff for Cohort B evaluations)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically-confirmed diagnosis of high-risk non-muscle-invasive (T1, high grade Ta and/or carcinoma

in situ [CIS]) transitional cell carcinoma of the bladder (mixed histology tumors allowed if transitional cell histology is predominant). 2. Fully resected disease at study entry (residual CIS acceptable). 3. BCG-unresponsive high-risk NMIBC after treatment with adequate BCG therapy. 4. Ineligible for radical cystectomy or refusal to undergo radical cystectomy. 5. Available tissue from a newly obtained core biopsy of a tumor lesion not previously irradiated.

4.2 Exclusion Criteria 1. Centrally assessed muscle-invasive, locally advanced nonresectable, or metastatic urothelial carcinoma (i.e., T2, T3, T4, and/or stage IV). 2. Centrally assessed concurrent extra-vesical (i.e., urethra, ureter, or renal pelvis) non-muscle invasive transitional cell carcinoma of the urothelium. 3. Active autoimmune disease that has required systemic treatment in the past 2 years or immunodeficiency requiring systemic immunosuppressive therapy. 4. Pregnant or breastfeeding, or expecting to conceive within the projected duration of the trial through 120 days after the last dose.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Drug Name: Pembrolizumab (MK-3475)
Trade Name: Keytruda **ATC Code:** L01FF02 **EMA Product Information:** https://www.ema.europa.eu/en/documents/product-information/keytruda-epar-product-information_en.pdf **Dosage/Frequency:** Pembrolizumab 200 mg every 3 weeks (Q3W). **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Up to 24 months (or 35 cycles), or until disease recurrence, progression, or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications. **Prohibited:** Systemic immunosuppressive therapies (e.g., high-dose corticosteroids), other systemic antineoplastic agents, prior therapy with an anti-PD-1/PD-L2 agent, and live vaccines within 30 days of planned start.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Scans, Biopsy Review
Baseline	Day 0	First Pembrolizumab Dose, Safety Baseline
Interim	Every 3 Weeks	Q3W IV Dosing, Safety Labs; Every 12 weeks: Cystoscopy, Cytology, Efficacy Assessment
End of Study	Up to Week 104	Final Assessment at 24 months or upon progression, IP Accountability, Transition to Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Cohort A:** Complete Response (CR) Rate of High-Risk Non-Muscle Invasive Bladder Cancer (NMIBC). (Defined as

the proportion of participants whose tumors are in response to treatment and who are free of high-risk NMIBC or worse per central pathology and radiology review). 2. **Cohort B:** 12-month Disease-Free Survival (DFS) Rate of High-Risk NMIBC. (Defined as the time from the first dose to the first occurrence of high-risk NMIBC or worse per central review). 3. **Cohorts A and B:** Number of Participants Who Experience an Adverse Event (AE).

7.2 Secondary & Exploratory Endpoints 1. **Cohort A:** CR Rate of Any Disease. 2. **Cohort A:** CR Rate of High-Risk NMIBC in Programmed Cell Death 1 (PDL-1) Positive Participants. 3. **Cohort A:** Duration of Response (DOR). 4. Progression-Free Survival (PFS) to muscle invasion, metastasis, or death. 5. Overall Survival (OS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Assessed continuously according to CTCAE v4.0 criteria (monitoring for immune-mediated AEs, fatigue, diarrhea, pruritus). An AE is defined as any untoward medical occurrence which does not necessarily have to have a causal relationship with this treatment. **Serious Adverse Events (SAEs):** Evaluated continuously and reported per standard 24-hour timelines. **Data Monitoring Committee (DMC):** External independent Data Monitoring Committee for safety and efficacy oversight.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: As-Treated/Safety Set (all patients receiving ≥ 1 dose of study drug); Efficacy analysis set (patients meeting disease definition criteria). **Sample Size Justification:** Powered to demonstrate a clinically meaningful CRR and DFS with a robust lower bound of the 95% confidence interval relative to historical benchmarks. **Handling of Missing Data:** Kaplan-Meier method applied for time-to-event outcomes (e.g., DOR, DFS, PFS); unconfirmed or missing responses are considered as non-responders.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring of source data, clinical assessments, and central pathology reviews. **Audit Trail:** Validated electronic data capture (EDC) systems with comprehensive audit trails, eCRF locking, and query resolution via data clarification forms.

11. Study Identifiers & Governance

Primary ID: MK-3475-057 **Registry Number:** NCT02625961 **Posting ID:** 79050007538655331 **Sponsor/Lead Organization:** Merck Sharp & Dohme Corp. (Merck) **Study Title:** KEYNOTE-057 **Official Regulatory Title:**

A Phase II Clinical Trial to Study the Efficacy and Safety of Pembrolizumab (MK-3475) and Pembrolizumab in Combination With Other Investigational Agents in Subjects With High Risk Non-muscle-Invasive Bladder Cancer (NMIBC) Unresponsive to Bacillus Calmette-Guerin (BCG) Therapy

12. Clinical Framework (NMIBC Specific)

Disease Area: Bladder Cancer **Primary Condition Preferred Name:** Malignant neoplasm of urinary bladder **Preferred UMLS Name:** High-risk non-muscle invasive urothelial carcinoma of bladder
Diagnosis Threshold: BCG-unresponsive high-risk NMIBC (CIS and/or papillary tumors) **Medical Methodology:** PD-1 Immune Checkpoint Inhibition **Optimization Target:** Complete Response and Bladder Preservation

13. Study Procedures & Methodology

Management Pathway: Pembrolizumab Monotherapy IV Infusion / Combination with Investigational Agents **Education Protocol (HCP):** Protocol-specific training for immune-mediated AE management
Education Protocol (Patient): Onsite training for AE symptom reporting and treatment adherence **Quality Audit Mechanism:** Independent central pathology and radiology review tracking

14. Observation & Follow-Up Schedule

Total Duration: 24 months (104 weeks) of treatment plus survival follow-up **Onsite Visit Cadence:** Every 3 weeks for treatment; tumor assessment every 12 weeks **Remote Monitoring Frequency:** Monthly follow-up for survival post-treatment progression
Checklist Review Interval: Every 12 weeks for cystoscopy and cytology assessment

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				